

BOSONIZATION OF FERMION CURRENTS IN TWO-DIMENSIONAL QUANTUM CHROMODYNAMICS

R.E. GAMBOA SARA VÍ¹, C.M. NAÓN² and F.A. SCHAPOSNIK¹

*Departamento de Física, Facultad de Ciencias Exactas, Universidad Nacional de La Plata,
C.C. No. 67, 1900 La Plata, Argentina*

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Bosonization rules for fermion currents in QCD₂ are constructed using a very simple approach within the path-integral framework, based on the existence of a gauge condition in which fermions can be decoupled.

Recently, there has been much progress in the understanding of two-dimensional non-abelian bosonization. On the one hand, Witten [1] has proposed a new bosonization scheme with the operator framework. On the other hand, Polyakov and Wiegmann [2] have established a very useful connection between non-abelian Goldstone fields and purely fermionic models, using the path-integral approach. In both investigations, the two-dimensional Wess–Zumino functional [3] plays a crucial role, thus providing a new link between models like QCD₂ or the Gross–Neveu one and more realistic four-dimensional theories.

Following these lines, many authors have studied in detail non-abelian bosonization. In particular, di Vecchia and Rossi [4], using functional techniques, analysed different regularisation schemes and the resulting free-fermion currents in terms of bosons (see also refs. [5–7]). Within the path-integral approach, González and Redlich [8] studied the modifications to the bosonization rules when interactions are considered.

Using recent results on fermion determinants [9–13], we study in this letter quantum chromodynamics with massless fermions in $d = 2$ space–time dimensions, giving explicitly the expression of fermion currents in terms of boson fields. This corresponds, in the path-integral framework, to bosonization rules for

interacting fermions, obtained in a very simple and systematic way; the method, a natural extension of the abelian analysis given in ref. [14], sheds light on the role of chiral anomalies in the game. The results, obtained in a rigorous and unambiguous way, represent an important step towards the complete solution of QCD₂. They agree with those obtained by González and Redlich [8] using a different approach, and reproduce, for the free-field case, the rules first given by Witten [1].

The clue in our approach is the use of a particular gauge-fixing: *the decoupling gauge-condition*, introduced in ref. [9], where the fermion determinant for QCD₂ was first computed. In this gauge, the fermions can be decoupled from gauge-fields becoming free, and a Wess–Zumino functional originated, as a result of a chiral change in the fermion variables. We think that the advantages of this gauge-condition have not been fully exploited in the literature, and for that reason we describe in some detail the formulation of the theory with this gauge-fixing. We then obtain fermion currents in a very simple way, using both the Schwinger method [15] and the mathematically more rigorous ξ -function prescription, recently developed in ref. [16]. From these results, the bosonization rules for the non-abelian case will be derived.

We start from the QCD₂ (euclidean) lagrangian

$$\mathcal{L} = \frac{1}{2} \text{tr} F_{\mu\nu}^2 + \bar{\psi}(i\partial + gA)\psi, \quad (1)$$

with $A_\mu = A_\mu^a t^a$ taking values in the Lie algebra of

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$SU(N)$, $\text{tr } t^a t^b = \frac{1}{2} \delta_{ab}$ and

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu - ig[A_\mu, A_\nu] \quad (2)$$

(we use the conventions of ref. [9]).

As we stated above, computations are particularly simple if one works in the decoupling gauge. Although the corresponding gauge-condition can be shown to be local [10], it is rather complicated, even in the simple $SU(2)$ case [13]. However, even without knowing explicitly this condition, it is possible to write any gauge field A_μ satisfying it, in the form

$$A = -(i/g) \bar{\phi} (\exp(\gamma_5 \phi)) \exp(-\gamma_5 \phi), \quad (3)$$

where $\phi(x)$ is a new field taking values in the Lie algebra of $SU(N)$, $\phi = \phi^a t^a$. As it is explained in refs. [10,12,13], there exists a local (non-linear) gauge condition such that any A_μ can be written as in eq. (3). Now, one can easily understand why this condition is so advantageous noting that the fermion determinant can be very simply evaluated by making a chiral rotation of the fermion fields; indeed, introducing new fermion variables:

$$\psi = \exp(\gamma_5 \phi) \chi, \quad \bar{\psi} = \bar{\chi} \exp(\gamma_5 \phi), \quad (4)$$

the computation of the fermion determinant reduces to the evaluation of the (quantum) jacobian associated to (4) [17,11] since, at the classical level, the fermion lagrangian becomes a free one:

$$\mathcal{L}_F = \bar{\psi}(i\bar{\partial} + gA)\psi = \bar{\chi}i\bar{\partial}\chi, \quad (5)$$

when one uses A_μ written as in (3). We have then

$$\begin{aligned} \det(i\bar{\partial} + gA) &= \int D\bar{\psi} D\psi \exp\left(-\int \mathcal{L}_F d^2x\right) \\ &= J(\phi) \int D\bar{\chi} D\chi \exp\left(-\int \bar{\chi}i\bar{\partial}\chi d^2x\right), \end{aligned} \quad (6)$$

or

$$W[A] \equiv \log(\det(i\bar{\partial} + gA)/\det i\bar{\partial}) = \log J(\phi). \quad (7)$$

The fact that $J(\phi) \neq 1$ is related to the (non-abelian) axial anomaly, and can be interpreted as a non-invariance of the fermion path-integral measure under chiral transformations [17].

The jacobian in (6) can be evaluated following the procedure developed in ref. [9] (see also ref. [11]). The answer is

$$\begin{aligned} W[A] &= \frac{ig}{2\pi} \text{Tr} \int d^2x \\ &\times \int_0^1 dt \gamma_5 \phi (\bar{\phi} A(x, t) - ig A(x, t) A(x, t)), \end{aligned} \quad (8)$$

where we have written

$$\begin{aligned} A_\mu^a(x, t) &= -(i/g) \text{Tr} [\gamma_\mu t^a \bar{\phi} (\exp(\gamma_5 \phi t)) \\ &\times \exp(-\gamma_5 \phi t)]. \end{aligned} \quad (9)$$

The parameter t takes values in the $[0,1]$ interval; it is introduced to build up the whole transformation (4) from infinitesimal ones. Remarkably enough, this parameter is the one at the origin of the extension from the S_2 sphere (compactified space-time) to the S_3^+ hemisphere in the Wess–Zumino functional (see below). Of course, any other $\phi(x, t)$ interpolating between $\phi(x)$ and 0 can be used instead of $\phi(x, t) = \phi(x) t$.

In order to write $W[A]$ in a compact form, showing explicitly the Wess–Zumino functional, we define new fields V_μ^t, A_μ^t through the identities (see also refs. [18, 19])

$$A_\mu(x, t) = V_\mu^t - \epsilon_{\mu\nu} A_\nu^t, \quad (10)$$

where

$$V_\mu^t = -(i/2g) [\partial_\mu \mathcal{U}^{1/2}(\phi, t), \mathcal{U}^{-1/2}(\phi, t)]_- \quad (11)$$

$$A_\mu^t = (1/2g) [\partial_\mu \mathcal{U}^{1/2}(\phi, t), \mathcal{U}^{-1/2}(\phi, t)]_+ \quad (12)$$

with

$$\mathcal{U}(\phi, t) = \exp(2t\phi). \quad (13)$$

Were we working in $d = 4$ dimensions, decomposition (10) would correspond to the usual vector-axial one. The fact that for $d = 2$, $\gamma_\mu \gamma_5 = i\epsilon_{\mu\nu} \gamma_\nu$, somehow blurs this interpretation. It is however clear at this point that the decoupling-gauge is the non-abelian generalization of the decoupling (Lorentz) gauge used to solve QED₂ [20] in the path-integral framework [21].

(Note that in this last case, $V_\mu^t = 0$ and $A_\mu^t = (t/g) \partial_\mu \phi$).

The following relations between A_μ^t and V_μ^t hold:

$$\partial_\mu V_\nu^t - \partial_\nu V_\mu^t - ig[V_\mu^t, V_\nu^t] + ig[A_\mu^t, A_\nu^t] = 0, \quad (14)$$

$$\partial_\mu \phi - ig[V_\mu^t, \phi] = g \partial A_\mu^t / \partial t, \quad (15)$$

$$\partial_\mu A_\nu^t - ig[V_\mu^t, A_\nu^t] = \partial_\nu A_\mu^t - ig[V_\nu^t, A_\mu^t]. \quad (16)$$

and can be used in order to write W in the form:

$$W[A] = \frac{1}{8\pi} \int d^2x \operatorname{tr}(\partial_\mu \mathcal{U}^{-1}(x, 1) \partial_\mu \mathcal{U}(x, 1)) \\ - \frac{i}{4\pi} \int_0^1 dt \int d^2x \operatorname{tr}(\epsilon_{\mu\nu} \mathcal{U}^{-1}(\phi, t) \partial_t \mathcal{U}(\phi, t) \\ \times \mathcal{U}^{-1}(\phi, t) \partial_\mu \mathcal{U}(\phi, t) \mathcal{U}^{-1}(\phi, t) \partial_\nu \mathcal{U}(\phi, t)) . \quad (17)$$

The second term on the RHS of eq. (17) is the two-dimensional version of the Wess–Zumino functional. Its presence in our approach has a transparent origin: the non-abelian chiral anomaly, responsible for the jacobian $J(\phi)$. Concerning the first term, it corresponds to the usual kinetic term in chiral field lagrangian (remember we are working in euclidean space; continuation to Minkowski space involves making $\phi \rightarrow i\phi$ and $\mathcal{U}(\phi) \rightarrow \mathcal{U}(i\phi) \in \mathrm{SU}(N)$).

Although W has been evaluated in the decoupling gauge, it can be explicitly written in a general form by noting that the substitution

$$\exp(\gamma_5 \phi) \rightarrow \exp[i\eta(x)] \exp(\gamma_5 \phi) \quad (18)$$

changes A_μ from the decoupling to a general gauge:

$$A(\phi) \rightarrow A(\phi, \eta) \\ = e^{i\eta} A(\phi) e^{-i\eta} + (i/g) e^{i\eta} \not{\partial} (e^{-i\eta}) . \quad (19)$$

From eq. (17), we see that a bosonic lagrangian has arisen from the decoupling of fermions. Being now free ones, one can trivially integrate the fermion degrees of freedom, leaving a purely bosonic theory.

Let us now establish the bosonization rules for fermion currents in this framework. We start from the (ill-defined) expression

$$J_\mu(x) = -(\delta/\delta A_\mu(x)) \log Z_F[A] = -\operatorname{tr} \gamma_\mu G(x, x), \quad (20)$$

where Z_F , the fermion generating functional, is given by

$$Z_F[A] = \int D\bar{\psi} D\psi \exp\left(-\int \bar{\psi} \not{D}(A) \psi d^2x\right),$$

$$\not{D}(A) = i\not{\partial} + g\not{A}, \quad (21)$$

and $G(x, y)$ is the Green function satisfying

$$\not{D}(A) G(x, y) = \delta(x - y). \quad (22)$$

Of course, eq. (20) needs a regularization in order to

handle the problem of multiplication of distributions at the same point. There is a very popular regularization technique created by Schwinger [15] where gauge invariance is preserved by the introduction of a phase factor:

$$J_\mu^a \operatorname{Sch}(x) = \lim_{y \rightarrow x} -\operatorname{tr} \left[t^a \gamma_\mu G(x, y) \right. \\ \left. \times \exp\left(ig \int_x^y A_\nu(z) dz_\nu\right) \right]. \quad (23)$$

An alternative prescription has been recently developed based on the ζ -function method [16]. It starts from the observation that one can handle the problem of regularization, once and for all, at the generating functional level. Indeed, Z being a determinant, i.e. a product of eigenvalues growing without bounds, it is not well defined. Now, in ref. [16] it is shown that

$$Z_{\operatorname{reg}}[A] = \exp[-(d/ds) \zeta(s; \not{D}(A))] |_{s=0}, \quad (24)$$

where $\zeta(s; D)$ is the generalized ζ -function for the operator D [22], then

$$J_\mu^a \operatorname{reg}(x) = -(\delta/\delta A_\mu^a(x)) \log Z_{\operatorname{reg}}[A] \quad (25)$$

is finite and can be taken as an unambiguous definition of the fermion current. It is important to stress that this approach, which can be rigorously justified from a mathematical viewpoint [16], leads automatically to a gauge-invariant answer. As it happens with the proper-time method, the explicit form of the Green function (at least for short distances) is needed. Again, it is precisely the use of the decoupling gauge condition that allows straightforwardly to get a closed answer:

$$G(x, y) = \exp[\gamma_5 \phi(x)] G_0(y - x) \exp[\gamma_5 \phi(y)], \quad (26)$$

where $G_0(z)$ is the free-fermion Green function

$$G_0(z) = (i/2\pi) \not{z}/z^2. \quad (27)$$

With G given by (26), one can easily get $J_\mu = J_\mu^a t^a$. Writing

$$J_\pm(x) = J_0(x) \pm iJ_1(x), \quad (28)$$

one gets (see ref. [16] for details of application of eq. (25))

$$J_\pm(x) = -(ig/4\pi) [\mathcal{U}^{\pm 1/2}, \partial_\pm \mathcal{U}^{\mp 1/2}]_+. \quad (29)$$

Using the Schwinger method, one also gets (29) after taking a symmetric limit. Within the ξ -function approach, the answer (29) is automatically (and rigorously) obtained just by computing a few Seeley coefficients [11].

Making use of relation (16) one can easily show that the current is covariantly conserved. Now, defining

$$D_{\pm} = \partial_{\pm} - ig[A_{\pm}, \] , \quad (30)$$

with

$$A_{\pm}(x) = A_0 \pm iA_1 , \quad (31)$$

one can rewrite (29) in the form

$$J_+(x) = -(ig/4\pi) \mathcal{U} D_+ \mathcal{U}^{-1} ,$$

$$J_-(x) = -(ig/4\pi) \mathcal{U}^{-1} D_- \mathcal{U} , \quad (32)$$

in complete agreement with the result of ref. [8], where a completely different approach was followed. Note that making $A_{\pm} = 0$ in (32), one gets the free fermion bosonization rules first derived by Witten [1].

It is interesting to note that eq. (29) (or eq. (32)) is nothing but the expression of $A_{\mu}^{t=1}$ in (euclidean) "light-cone" coordinates. This is no coincidence and in fact, this relation, also true in the abelian case, has been exploited in ref. [14] in order to derive the abelian bosonization rules in the path-integral framework. Moreover, it confirms Swieca's [23] observation on the importance of the relation $J_{\mu} \sim A_{\mu}$ in the understanding of bosonization.

We then conclude by stating the bosonization rules for fermions interacting with non-abelian gauge fields: the fermion lagrangian is equivalent to a bosonic one defined by eq. (17) and the fermion currents are written in terms of bosonic fields as in eq. (32).

In order to complete the analysis of the QCD₂ model, one has to express the $F_{\mu\nu}^2$ in (1) in terms of $\mathcal{U}$ fields and also write the Faddeev–Popov determinant for the decoupling gauge. Although this last calculation may become complicated, we think the whole approach is very useful in the sense it provides a systematic way of analysing abelian and non-abelian mod-

els in a unique bosonization scheme, also applicable to the massive case [24]. Moreover, it sheds light on the role of the (non-abelian) chiral anomaly and the appearance of the Wess–Zumino functional, an aspect from which one might gain insights into how low-energy non-linear sigma models arise from QCD in four-dimensional space.

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